

Prompt 3-1: using the finite difference method

Solve this 3D elasticity problem from scratch using the finite difference method.

Problem details: A 3D domain $[0,1]\text{m} \times [0,1]\text{m} \times [0,1]\text{m}$ ($x \times y \times z$) representing a 3D cube. Linear elasticity, elastic modulus $E=3 \text{ MPa}$, Poisson's ratio $\nu=0.4$, no body forces.

Boundary conditions:

- At $z=0$ (all x, y in the domain): $u_x=0, u_y=0, u_z=0$; all rotations constrained (fully fixed).
- At $z=1$ ($x \in [0,0.5]\text{m}, y \in [0,0.5]\text{m}$ in the domain): $u_z=0.1\text{m}$.
- All other boundaries: traction-free.

Calculations required: u_x, u_y, u_z displacement fields; strain fields; stress fields (3D Visualization of the Results).

Programming language: Python 3.10.

Permitted libraries: NumPy 2.1.1, SciPy 1.14.1, JAX 0.4.30, Matplotlib 3.9.2.